

The Impact of School Weapon Threats on Student Physical Health

An Analysis of BMI Percentiles via ANOVA and Multiple Regression

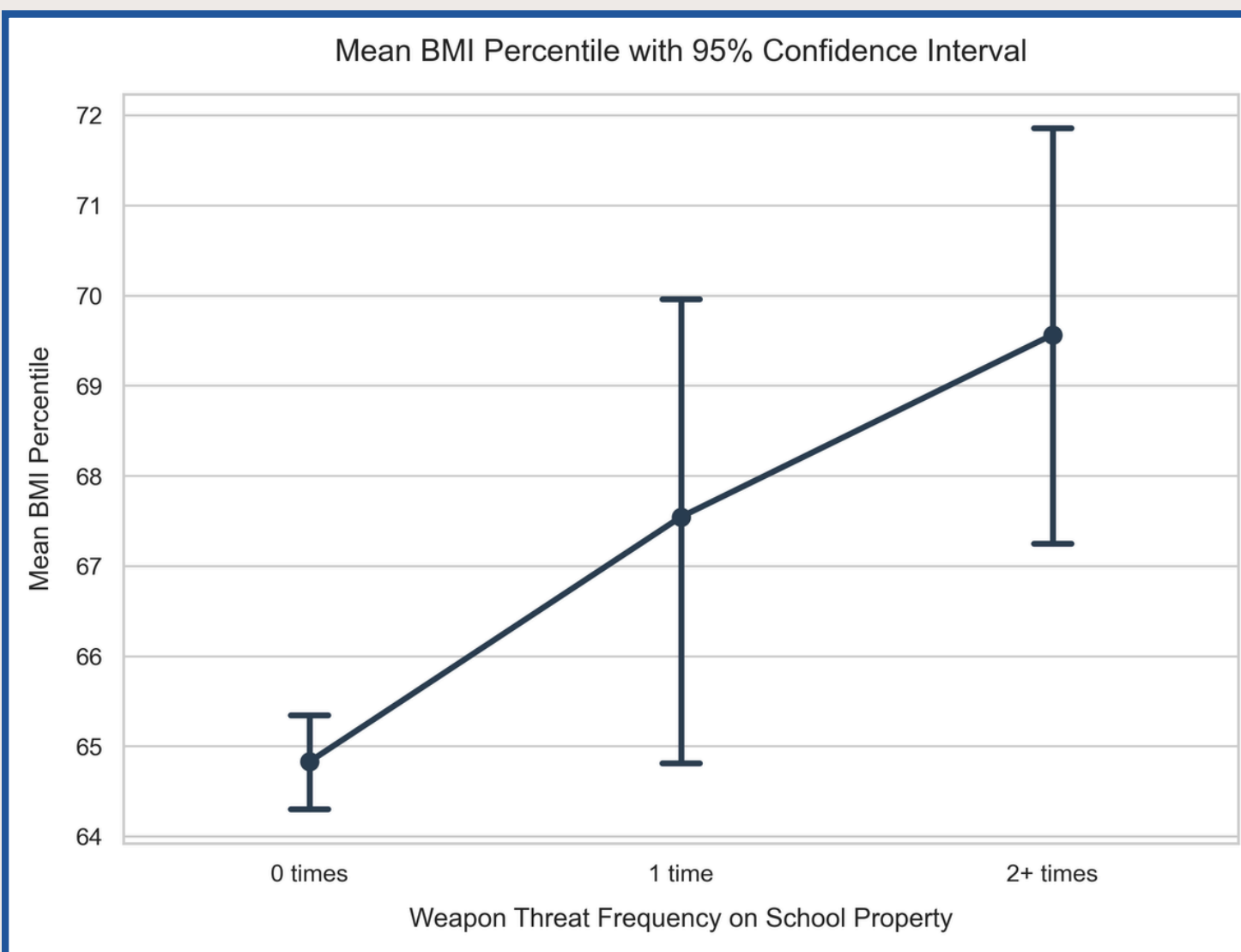
Research Question & Methodology

Research Question: Does the frequency of facing weapon threats at school significantly impact a student's BMI, even after accounting for their exercise habits?

Data & Methodology:

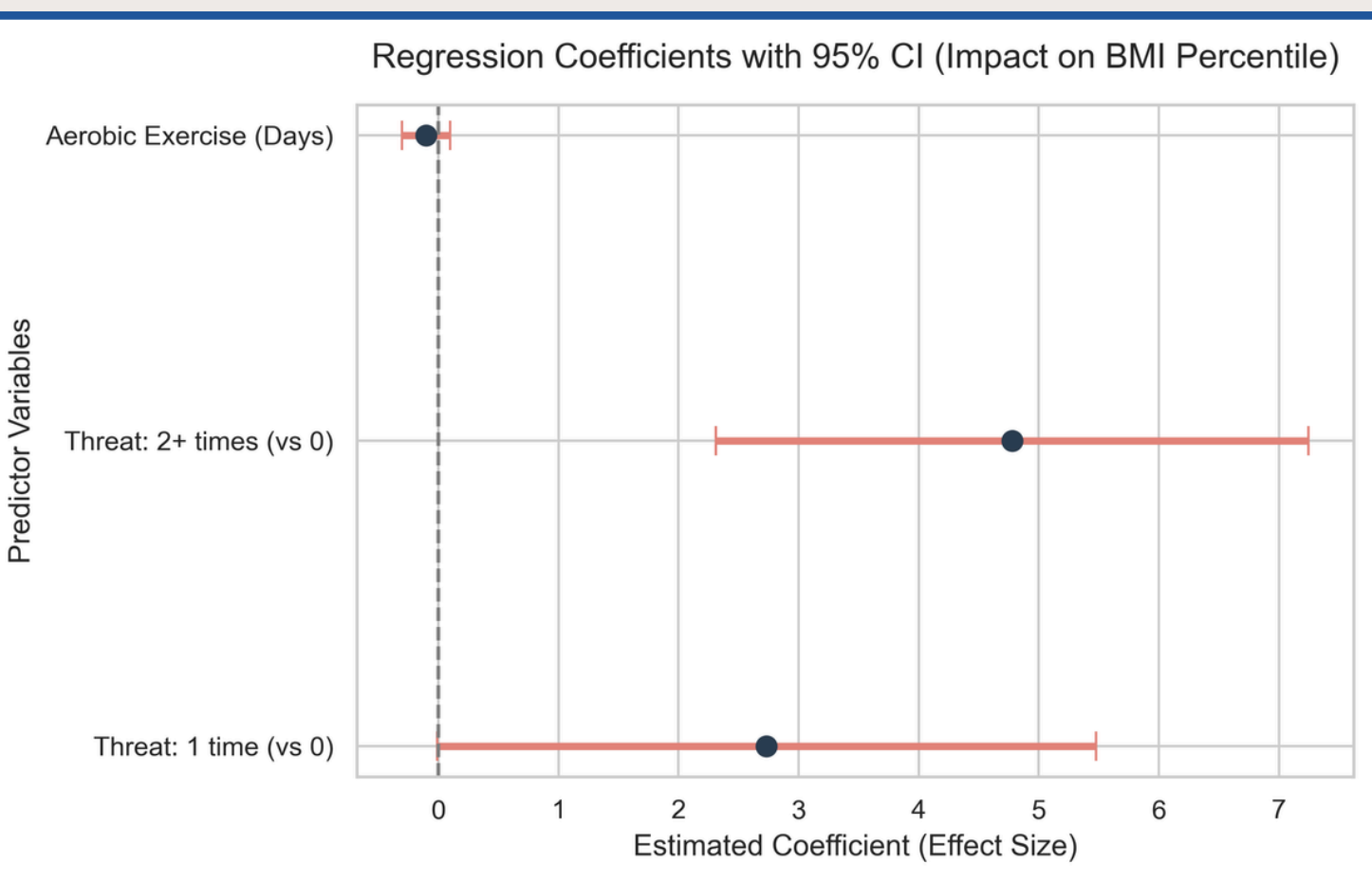
- **Dataset:** YRBS 2007
- **Dependent Var (Y):** BMI Percentile
- **Main Predictor (X1):** Weapon Threat Frequency (0, 1, 2+ times)
- **Control Var (X2):** Aerobic Exercise (Days)
- **Methods:** One-Way ANOVA & OLS Regression

Key Findings & Visual Evidence



Clear Upward Trend: Students facing frequent threats show significantly higher mean BMI percentiles.

Significant Difference: The 95% Confidence Intervals do not overlap, confirming the ANOVA results ($p < 0.05$).



Robust Threat Effect: Even after controlling for exercise, weapon threats remain a positive and significant predictor of higher BMI (Red line > 0).

Control Variable Check: Aerobic exercise properly shows a negative correlation with BMI.

Statistical Summary & Conclusion

ANOVA: $P\text{-value} < 0.05 \rightarrow$ Reject H_0

Regression: Threat Effect $P\text{-value} < 0.05 \rightarrow$ Significant

Control Check: Aerobic Exercise $\rightarrow$ Lowers BMI (Expected)

Higher Threat = Higher BMI: The frequency of school weapon threats positively correlates with student BMI percentiles.

Independent Effect: This psychological/environmental impact remains significant even when controlling for students' aerobic exercise habits.